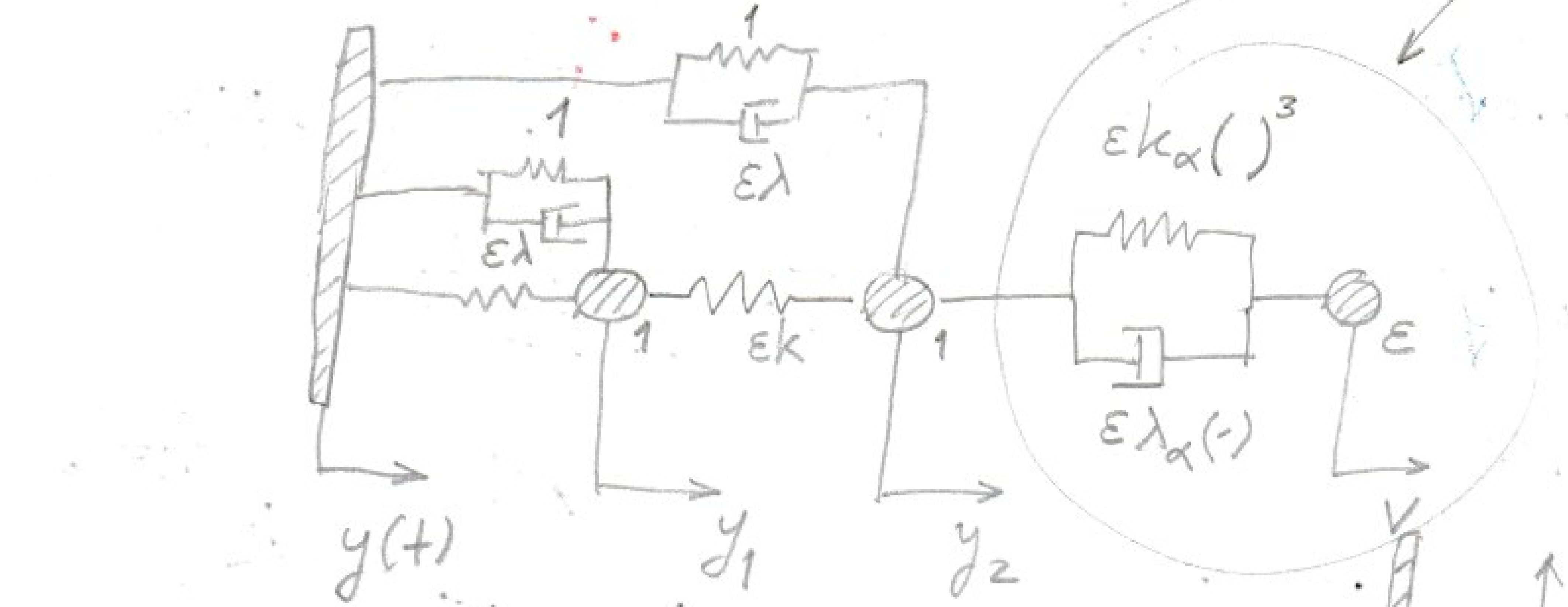
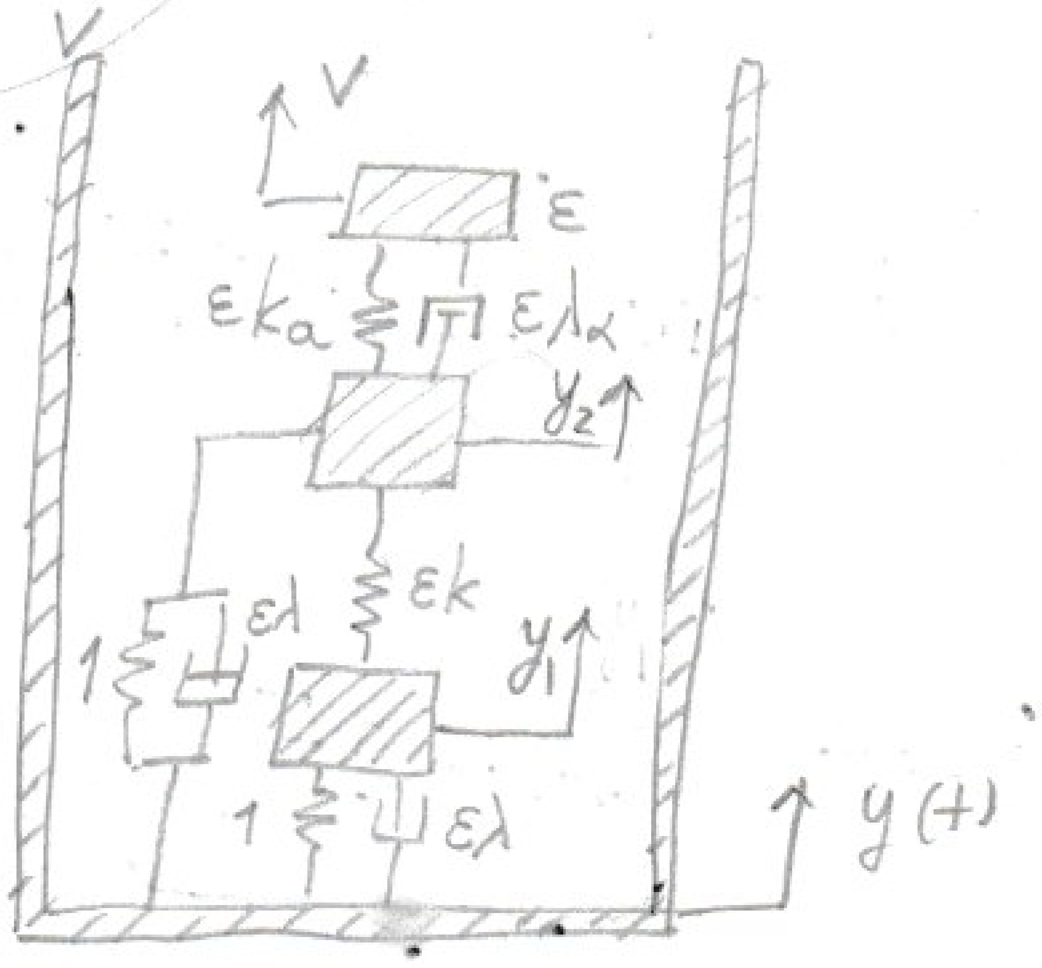
Case of base excitation

At a later stage replace this hardening NL with geometric NL (page 1)

Must show through existence of hysteresis loop in SIM!





$$\ddot y_1 + (y_1-y) + \varepsilon\lambda_\alpha(\dot y_1-\dot y) + \varepsilon k(y_1-y_2) = 0$$

$$\ddot y_2 + (y_2-y) + \varepsilon\lambda(\dot y_2-\dot y) - \varepsilon k(y_1-y_2) + \varepsilon k_\alpha(y_2-v)^3 + \varepsilon\lambda_\alpha(\dot y_2-\dot v) = 0$$

$$\varepsilon\ddot v - \varepsilon k_\alpha(y_2-v)^3 - \varepsilon\lambda_\alpha(\dot y_2-\dot v) = 0$$

$$\left.\begin{array}{l} \text{Let } z_1 = y_1 - y,\ z_2 = y_2 - y,\ z_3 = v - y \Rightarrow \\ \Rightarrow y_1 = z_1 + y,\ y_2 = z_2 + y,\ v = z_3 + y \end{array}\right\} \Rightarrow$$

$$\ddot z_1 + \ddot y + z_1 + \varepsilon\lambda\dot z_1 + \varepsilon k(z_1 + \cancel{y} - z_2 - \cancel{y}) + \varepsilon\lambda(\dot z_1 - \dot z_2) = 0 \Rightarrow$$

$$\boxed{\begin{array}{lr} \Rightarrow \ddot z_1 + z_1 + \varepsilon\lambda\dot z_1 + \varepsilon k(z_1 - z_2) + \varepsilon\lambda_\alpha(\dot z_1 - \dot z_2) = -\ddot y & (20) \\ \ddot z_2 + z_2 + \varepsilon\lambda\dot z_2 - \varepsilon k(z_1 - z_2) + \varepsilon k_\alpha(z_2 - z_3)^3 + \varepsilon\lambda_\alpha(\dot z_2 - \dot z_3) = -\ddot y & (21) \\ \varepsilon\ddot z_3 - \varepsilon k_\alpha(z_2 - z_3)^3 - \varepsilon\lambda_\alpha(\dot z_2 - \dot z_3) = -\varepsilon\ddot y & (22) \end{array}}$$